

Project Initialization and Planning Phase

Date	6 July 2026
Team ID	xxxxxxx
Project Title	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to revolutionize online learning support by implementing advanced NLP techniques (BiLSTM & BERT) to detect a student's emotional state from text and dynamically generate empathetic, personalized learning strategies using Generative AI.
Scope	The project comprehensively assesses and enhances the student learning experience, providing a unified web application that tracks academic struggles and emotional wellbeing in real-time.
Problem Statement	
Description	Addressing the lack of emotional intelligence in traditional online learning and standard AI tools, which often leave frustrated or anxious students overwhelmed with generic, robotic answers.

Impact	Solving these issues will result in improved student retention, reduced academic anxiety, and an overall enhancement in the learning process, contributing to greater student confidence and success.
Proposed Solution	
Approach	Employing deep learning text-classification models (BiLSTM, BERT) to analyze and predict emotional states, paired with the Gemini API to create a dynamic, empathetic learning support system.
Key Features	- Implementation of machine learning-based emotion classification models. - Integration of Google Gemini API for empathetic, context-aware advice generation. - Real-time Analytics Dashboard for tracking emotional journeys over time.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard multi-core CPU (Optional T4 GPU for model training)
Memory	RAM specifications	8 GB RAM (minimum)
Storage	Disk space for data, models, and logs	20 GB SSD
Software		
Frameworks	Python frameworks	Streamlit
Libraries	Additional libraries	tensorflow, transformers, google-genai, pandas, plotly, scikit-learn

Development Environment	IDE	VS Code, Jupyter Notebook
Data		
Data	Source, size, format	Text Emotion datasets (e.g., Kaggle GoEmotions), CSV format